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Sustainable development model of EU cities compliant with UN settings

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Abstract: Nowadays the globally accepted UN concept of the sustainable development (SD) is transferred to the city level, including small and medium-sized cities. The implementation of the SD settings requires regular measurement of development progress to monitor the achieved level in statics and dynamics, as well as to make strategic decisions for the next period. The existing urban SD indicator systems and indexes are poorly usable for the monitoring of the specific city. A mathematical model of the city SD is created, which is compliant with the UN vision, is usable for any city, reflects the level of quality of life achieved, includes a limited number of indicators related to municipal capacity and activities for increasing the SD level, ensures the most possible objective selection and proportions of specific indicators in the model. The benchmarking algorithms and mathematical modelling procedures are applied to develop an appropriate model and methodology for measuring the SD level achieved. Mathematical calculation instead of traditional voluntarily preference of some indicators, the reasonable combination of accuracy, stability and simplicity of the model are strong advantages for the practical applications. The performed stability test confirms that annual calibration of the model is not necessary. The index will help municipalities in the rational use of their usually limited resources and provide possibility of comparative assessment of the proposed development projects in the different sectors. As a pilot project, values of the index were calculated for several Latvian cities.

Keywords: Agenda 2030; sustainable urban development; measuring development; local governance; benchmarking modelling; key performance indicators.

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1. Introduction

A sustainable development (SD) concept was institutionalized by the UN as the core of the global development in the 21 century [1]. It was noted from very beginning that limitation to "… environmental issues only would have been a grave mistake. The environment does not exist as a sphere separate from human actions, ambitions, and needs." [2]; the concept is based on three independent and mutually reinforcing pillars – economic, social and environmental. Elaborating the global development program *Agenda 2030* [3] the concept has been detailed by setting 17 sustainable development goals (SDG) and 169 targets for the period up to 2030.

Knowledge, which certainly is the main resource of the SD, has a necessity and tendency to concentrate; it supports existing knowledge to create and to attract new one: "knowledge, professional skills, creativity and technological inventions have never been evenly distributed in space" [4]. The knowledge-based SD process has become a catalyst for the worldwide urbanization; more and more knowledge, economic and social resources and activities are concentrated in cities, including small and medium-sized ones (see, e.g., [5]). "The 19th century was a century of empires. The 20th century was a century of nation states. The 21st century will be a century of cities." [6]. The concept of the

SD, both the benefits and the problems, is actually transferred at the city level: "In the 21st century cities are at the centre of all key demographic, environmental and socio-economic trends." [7].

The implementation of the SD settings requires regular measurement of development progress to monitor the achieved level in statics and dynamics, as well as to make strategic decisions for the next period. In order to reflect the situation in various aspects, the UN SDG recommendations offer 242 narrow-profile indicators; numerical values of 100 SD indicators are available in the most comprehensive Eurostat database [ ].

However, the direct measurement of many individual indicators cannot show the overall progress of the multi-dimensional SD program due to complicated crosslinks, integrity and interplay of dimensions. To assess the progress of SD, a composite index is needed, which covers all dimensions of SD; "Indicators corresponding to the future SDGs are most important for monitoring future progress, but they will need to be complemented by composite indices of sustainable development progress." [8].

Experts have created a number of indicator systems and integrated indexes to assess the urban SD; they contain a large number of parameters chosen by developers in unreasonable proportions (see, e.g., [9-11]). The set goals are quite similar although they are defined differently [12]: sustainable city [13, 14], smart city [15-17], liveable city [18, 19], eco-city [20]. Unfortunately, the mutual relationships and inter-impact of the parameters and development dimensions are not taken into account.

The proposed indexes and the methodologies of their creation are poorly usable for the data-driven planning and implementation of the SD of the specific city. The sets of indicators: (1) are selected voluntarily, it is not justified whether exactly they are the priority for the city SD; (2) include national level indicators, which don't relate to the functionality of municipalities; (3) include indicators specific to large cities only; (4) are not consistent with the globally accepted UN SD paradigm. The proportions of the indicators in the calculated complex index are voluntarily chosen, integrity and interplay of indicators is not taken into account in the calculations. The obtained index is not compliant with the UN SD vision.

COVID-19 pandemic has significantly damaged urban development, "the overall global average liveability score has fallen by seven points" [19]. The follow-up long recovery period significantly increases the importance of sustainability in global, national and city policy-making, both in the short and long term. Whereas the pandemic is still going on, there is a risk that the long-term SD trend will not be supported by the short-term post-COVID recovery activities.

The aim of the study is to develop a mathematical model of the city sustainable, the methodology for calculating the achieved level of the SD and the corresponding City Sustainable Development Index (CSDI) which:

- are usable for any city, including small and medium-sized ones:
  - ✓ correspond to the functionality of the city municipality – regulation and administration in the sectors of local government autonomous functions and delegated public administration tasks, provision of services financed or co-financed by the budget and paid services, maintenance and development of public infrastructure, promotion of the civil society and business development;
  - ✓ the municipality of the small city also can manage and improve performance results;
- reflect the level of quality of life of the inhabitants of the respective city;
- are consistent with the UN SD vision;
- include a limited number of the key performance indicators (KPIs);
- ensure the most possible objective selection and proportions of KPIs in the model.

In chapter 2 the methodology of the CSDI development is described. Chapter 3 is devoted to the linear and non-linear modelling process. Chapter 4 discusses the results obtained, while chapter 5 contains conclusions and recommendations for the introduction of the CSDI in practice of civil servants, experts and politicians.

2. The methodology

The direct theoretical calculation of the SD model's function and the subsequent measurement of the progress of the SD are impossible. Therefore, to achieve the aim, the adapted data mining methods were used, which are well suited to discover and generate the knowledge about the existing regularities in various data sets. Algorithms of the regression analysis and benchmarking were applied to explore cause and effect relationships, to identify the best performing leaders, and how the currently less successful and lagging behind could progress faster. The authors' previous experience in using modelling procedures to simulate various socio-economic processes confirms the perspective of such an approach [21, 22].

The benchmarking modelling means search of the mathematical model that is based on observations of specific cases (see, e.g., [23]). For our task the modelling determines the relationships between the indicators, which describe some aspects of the SD, as independent (input) variables (predictors,) and some SD indicator, which is fully compliant with the UN settings, as the dependent (target) variable; its increase reflects the progress achieved. If the impact of external factors on all cases is similar, the benchmarking provides comparative assessment of the SD level; this way we disclosed the impact of various indicators on SD level and extracted the most significant ones (key performance indicators (KPI)), reducing a large number of predictors.

These algorithms can be applied using already known complete data sets; unfortunately, they are not available at the urban level. To overcome this shortcoming, we had to use a detour for the computations, basing the benchmarking modelling on the statistical data of the EU27 country level, because of the homogeneous environment with the EU cities. The available EU27 level data sets include indicators on countries of different sizes with different populations, including relatively small countries with small populations (e.g., Luxembourg, Malta). There are indicators that are neutral in relation to the size of the area covered (e.g., percentage of the total number of population, enterprises, area, etc.). The linking of indicators to the concrete country may be in no way defined and/or used in the development of the mathematical model. In general, this means that the mathematical expression found can also be used to estimate the SD of cities.

The dependent (target) variable for the data mining should be fully consistent with the SD vision of the *Agenda 2030*, globally politically accepted by the UN: balance and integrity of economic, social and environmental dimensions. A suitable tool – the Advanced Human Development Index (AHDI) – has been created by supplementing the globally accepted Human Development Index (HDI) with multidimensional Environmental Performance Index (EPI) according the *Agenda 2030* vision [24]; the methodology for its elaboration ensures the necessary consistency. We used the AHDI as the target variable for modelling.

Direct transfer of global headline indicators (independent variables of the AHDI) to the urban level is neither possible nor targeted, as they cover national economic, regulatory and other aspects, which are determined and widely used at a high political level. They cannot be defined at the level of an individual city (e.g., Gross National Income, Purchasing Power Standard, schooling years, etc.) as well as cannot be influenced in any way by the municipality. For the motivated development of the city, the input data set should include indicators at least in part related to the competence of the city municipality so enabling activities to improve (see also [25]).

In fact, the set of the SDGs already detail the UN's understanding on the SD format, while related indicators show progress towards achievement of each SDG as the result of performed activities. This means that the selected indicators should correspond to one (or several) SDGs, so preserving the compliance with the UN vision.

There is a stand-alone SDG11 in the Agenda 2030 related to the cities and urban development. Several of its targets related to housing, basic public services, healthcare, pollution are more detailed in SDG3, SDG5, SDG6, SDG7, SDG12, SDG13 and SDG15. There is also recognition of the cross-cutting nature of urban issues, which are interlinked "with a number of other SDGs, including SDG6, SDG7, SDG8, SDG9, SDG12, SDG15, and SDG17 among others" [7]. We also considered SDG4 (Quality education) and SDG16 (Peace, justice and strong institutions), related to "human resource development and capacity building", as important for the urban level.

In turn, we did not include in the study several SDGs as inconsistent with functionality of municipalities:

- SDG14 does not correspond to the city level;
- poverty and hunger (SDG1, SDG2) in the UN comprehension are not relevant in the EU27;
- the key target of the SDG10 (Reducing inequality) is achievable only as a result of national political settings and decisions – minimum income level, minimum wage level, distribution of benefits through the tax system, etc.

So 13 of 17 SDGs relate fully or in part to the functionality of the city council and municipality. This means that the municipality should work simultaneously on many different action lines to achieve progress in SD. It is absolutely clear that no municipality can perform all of them, even in part, due both for the insufficient capacity to cover the whole very wide spectrum and inability to invest simultaneously in all action lines. The prioritization of the action lines is an essential, sometimes even critical factor for many municipalities.

To obtain relatively abundant data sets at EU27 country level, we used as the independent variables for the data mining and modelling not only statistical data from the Eurostat, but also data from other databases, i.e., The Organisation for Economic Cooperation and Development, The World Bank, The Council of European Municipalities and Regions, The Digital Agenda Scoreboard, The European Innovation Scoreboard, Eurobarometer. These data are processed according to unified methodologies and regularly updated; their quality is sufficient for modelling.

Several sequential steps were required to create the model (fig. 1).

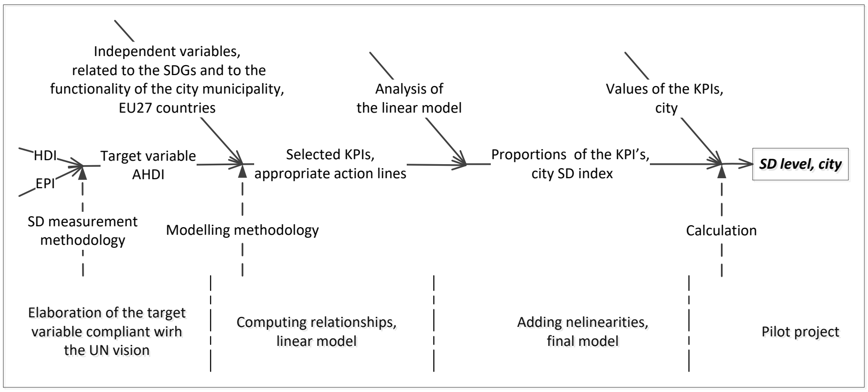

Figure 1. Creation of the model. Developed by authors.

The modelled AHDIm would be defined as a multi-parameter function (f) of the set (Π) of the *n* KPIs (p_n):

$$\text{AHDIm} = f(\Pi) = f\,\{p_1, p_2, \ldots, p_n\} \qquad (1)$$

The KPIs in the modelling process were computed from the gathered broad set of indicators. To achieve the objective selection of the KPIs, as many quantitative indicators (which relate to the municipalities) as accessible were selected and tested for each SDG (49 indicators in total, see Appendix). The achieved level of compliance of the model's target values AHDIm to the real values AHDI (mutual correlation) served as a quality criterion of the modelling procedure. The determination coefficient R^2 was used as the unit of correlation during the modelling.

The found KPIs for EU countries are not different from those for EU cities, due to defined principles for selecting indicators and an analogue SD treatment. The found mathematical expression (resp., the specific weights of the KPIs too) can be transposed at the urban level and used for the calculation of the CSDI, resp., the SD level of the city.

3. Modelling process

3.1. Linear modelling

The choice of the optimal modelling algorithm tools is determined by the requirements set by the task to be solved: (1) the mode and tool of modelling should allow easy repeatability of the modelling if data and/or indicators change; (2) the model should be implementable and adaptable to specific city by a person with average programming skills.

We started modelling by using the linear regression algorithm, because (1) it is mathematically the simplest method, (2) there is a simple and clear interpretation of the model, and (3) model can be easily calculated without special knowledge in mathematics and programming. Initial *ad hoc* experiments in creating a linear regression model showed good results, so it was decided to use a linear modelling. Post-modelling analysis will show the need and purposefulness to use nonlinearities to obtain a stronger causal relationship.

By the linear algorithm the benchmarking model (1) is expanded as the linear mathematical expression:

$$AHDIm = \alpha + \beta_1 * p_1 + \beta_2 * p_2 + \ldots + \beta_n * p_n \quad (2)$$

where:

α – the intercept;

β_n – modelled proportion of the KPI (p_n) in the AHDIm.

Several general and specialized programming languages are suitable for our task. Among them is the language R, which is specialized for data analysis; it has a number of advantages that are important to our study: (1) open access to the most popular operating systems; (2) a qualitative connected open Integrated Development Environment (IDE) RStudio; (3) an interpretation mode that speeds up the program development; (4) a lot of open access external libraries for data analysis and display; (5) the syntax of the language is simple and quick to learn.

Language R allows an interactive execution of the commands *on the fly* and an immediate display of the result; this speeds up the *ad hoc* analysis of data. It is possible to create complex data analysis programs. RStudio is a user-friendly environment for the development that allows interacting as well as developing and operating complex applications. For these reasons, we chose to use the well-developed language for data analysis and modelling R and the associated IDE RStudio. The general algorithms and the standard software package needed to be adapted and supplemented to perform our specific tasks.

The gathered indicators should associate with one of the 13 SDGs related to city-level performance. To achieve the most objective mathematical selection of KPIs by the modelling process, as many quantitative indicators as accessible were gathered; the full list of the indicators is in the Appendix). There were created 13 separate groups of indicators related to one SDG; each group x consists of a series of indicators (Nx indicators Gx.Kx that are associated with each SDGx, see fig. 2), and used for modelling as independent variables.

To obtain reliable modelling results based on the causal relationships between the P independent variables and the dependent one and to exclude individual deviations, T>>P is required, where T – the number of cases (27 countries in our study). The stronger this inequality, the more accurate the causal relationship (from the point of view of general causality) is created. If T≤ P, we could certainly find even several relationships that perfectly cover all T data points, but without the possibility of further generalization (what we need to transfer result to cities).

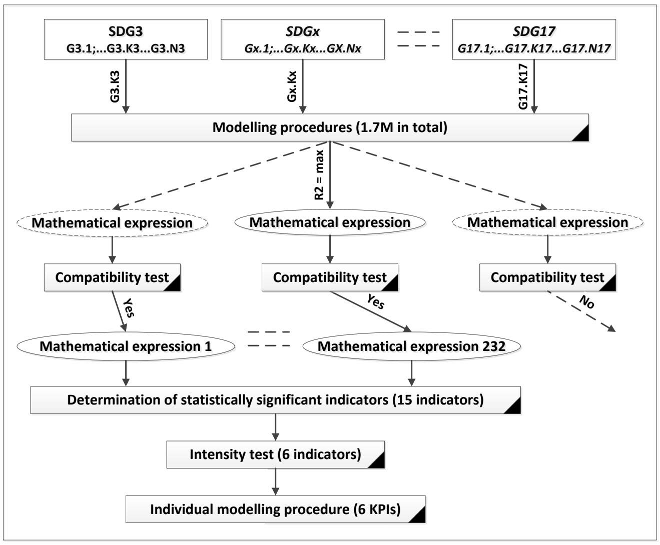

Figure 2. Linear modelling process. Developed by authors.

However, according to the UN's vision of a balanced SD, all SDGs should be equally represented in the search for KPIs in the modelling process. Together, this means that only one of indicators Gx.Kx from the set of indicators of each SDGx can be used as the predictor in each modelling procedure.

Obviously, the highest obtained efficiency of the SD process will be achieved if all terms in expression (2) are compatible, and the progress of all used predictors provides a positive effect on the progress of the target variable AHDI. To disclose such *good* combinations of predictors, it is necessary to model all possible combinations, in which a single indicator from each SDGx indicators' set is used as the predictor; more than 1.7 million modelling procedures were performed in total. Each modelling procedure was supplemented with a compatibility test to check the mutual compatibility of the specific indicators' combination. 232 *good* combinations were identified; they were analyzed for determination of statistically significant indicators.

15 indicators of the 49 ones were found to be statistically significant at least in one modelling procedure using even low threshold for the evaluation of predictors' significance: the probability for rejection of the null hypothesis (p-value) is less than 0.1. These indicators divided into two drastically diverse groups by the intensity of their inclusions in the *good* combinations: eleven indicators were represented less than 10 times each, while six indicators more than 25 times each (table 1; see indicator numbering in the Appendix).

Using only these six indicators as predictors we performed individual modelling procedure, creating the mathematical expression of the 6-predictors' linear model AHDIm1 (3):

$$\mathrm{AHDIm1} = 0.6527891 + 0.0012563^*(\mathrm{G3.1}) - 0.001596^*(\mathrm{G6.2}) - 0.0019518^*(\mathrm{G7.2}) + \\ +0.0013329^*(\mathrm{G9.2}) + 0.0006729^*(\mathrm{G12.1}) + 0.0010646^*(\mathrm{G16.2}) \quad (3)$$

The modelled linear expression (3) reflects reality with a high accuracy. The determination coefficient R^2 indicates very strong correlation between actual AHDI and modelled AHDIm1 (fig. 3), while the p-value is extremely low (table 2). The adjusted determination coefficient R^2_{adj} shows that more than 90% of variability in the data is explained by the model. All predictors are of high statistical significance; the small p-values of the predictors show their decisive role in the model's regularity. It clearly means that they are the searched KPIs; action lines, which lead to progress in these KPIs, can be recommended to municipalities to increase sustainability of their cities.

Table 1. Key performance indicators.

No	Key Performance Indicators
G3.1	Share of people with good or very good perceived health, % of population aged 16 or over
G6.2	Population not having indoor flushing toilet for the sole use, % of population
G7.2	Share of population unable to keep home adequately warm by poverty status, %
G9.2	Percentage of enterprises who have ERP software package to share information between different functional areas (10+ employees)
G12.1	Recycling rate of municipal waste, % of generated
G16.2	Perceived independence of the justice system, % of population perceived as very good or fairly good

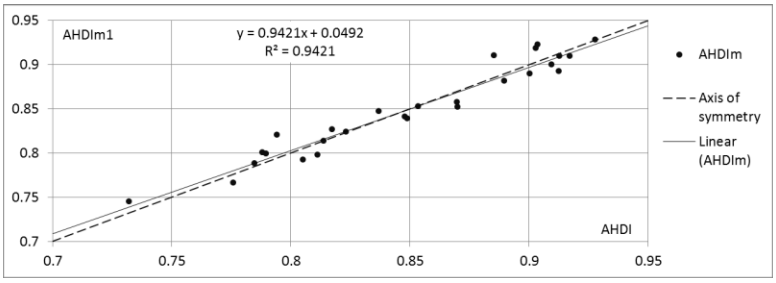

Figure 3. Regularity of the linear model AHDIm1 vs AHDI. Developed by authors.

Table 2. Numerical characteristics of the linear model AHDIm1 and the nonlinear model AHDIm.

Model	R^2	R^2_{adj}	p-value	Max residual	Residual standard error	Number of residuals	
						>2.5%	1.5-2.5%
AHDIm1 (3)	0.9421	0.9247	2.502e-11	0.0258894	0.01472	2	8
AHDIm (8)	0.9637	0.9622	<2.2e-16	0.0197030	0.01042	0	3

3.2. Non-linear modelling

Despite the excellent numerical characteristics of the model, detailed analysis shows also the possibilities for its further improvement. The R diagnostic plot (fig. 4a) shows that residuals are not quite evenly spread around a horizontal line (especially at high fitted values). It is an indication that the linear model doesn't fully capture the existing non-linear causal relationships. Also the plot (fig. 4b) identifies the data point 23 that lies behind the threshold, so called *Cook's distance*. This data point is not an outlier; nevertheless in the model (3) it can become influential against a general regularity. A reduction of residuals would also be desirable. Together, these point to a challenge to add some non-linearity in the model in order to improve further the compliance of the modelled data with the actual ones. This was done both at the level of individual predictors and at the level of the mathematical expression of the model.

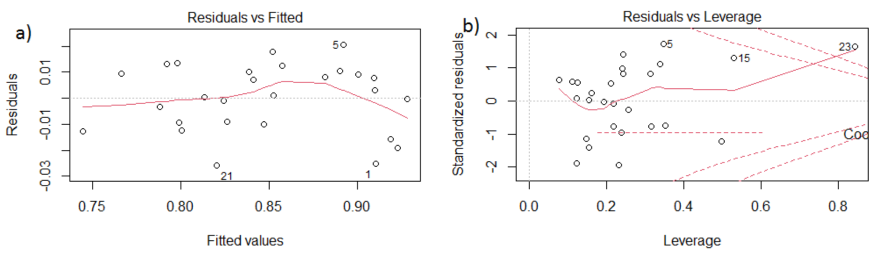

Figure 4. Residual characteristics of the linear model. Developed by authors.

To process the predictors' level, we checked the individual causal relationships between each predictor and the target variable AHDI. The real impact of each individual predictor on the AHDI is, of course, different from the individual regularity (e.g., due to some mutual impact of predictors). Nevertheless, the qualitative differences between the six individual regularities provide some comparative indication. Five of the six regularities show weaker or stronger proportionality. However, the sixth regularity (G6.2 versus AHDI) is obviously very far from linear (fig. 5a); this, of course, reduces the quality of the linear model.

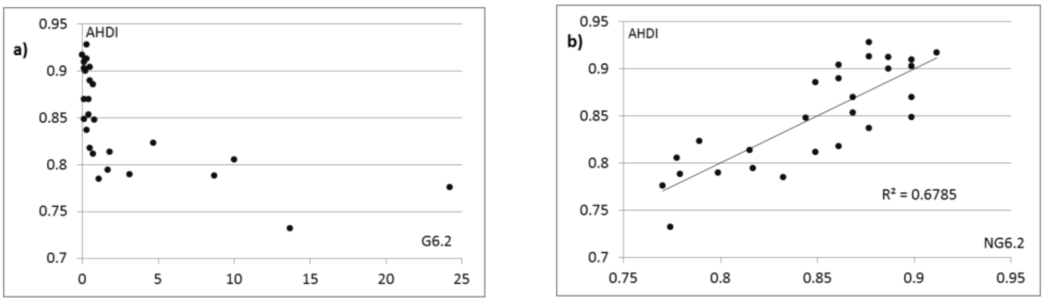

Figure 5. Regularity G6.2. vs AHDI (a) and regularity NG6.2 vs AHDI (b). Developed by authors.

By means of the *R studio* function *NLS* we modelled several following functions to transform the G6.2 data set into a new data set NG6.2, achieving stronger linear relationship between NG6.2 and AHDI:

$$\text{AHDI} \sim a + b * (G6.2)^c$$

$$\text{AHDI} \sim a * \ln(b * G6.2) + \text{c}$$

$$\text{AHDI} \sim \frac{a}{G6.2+c} + b \qquad (4)$$

The strongest correlation (fig. 5b), using the correlation *correl(NG6.2:AHDI)* as the quality criterion, was achieved by modelling the inverse proportionality expression (4), obtaining the NG6.2 expression (5):

$$\text{NG6.2} = 0.13606/(\text{G6.2} + 0.91652) + 0.76488 \qquad (5)$$

By repeating the linear modelling, we obtained corrected AHDI model M6:

$$\text{M6} = 0.3781410 + 0.0005767*\text{G3.1} + 0.3869453*\text{NG6.2} - 0.0017305*\text{G7.2} + + 0.0011321*\text{G9.2} + 0.0006888*\text{G12.1} + 0.0009180*\text{G16.2} \qquad (6)$$

Scatterplot (fig. 3) shows that the linear trendline of the data points is slightly skewed with respect to the axis of symmetry. As a result, smaller fitted values AHDIm generally are slightly above corresponding AHDI values, while large fitted values are below. Such shifts indicate that the sigmoidal function is best suited for improvement of the model. We modelled several S-shaped functions:

$$\text{AHDI} \sim \alpha + \beta(\text{a} * \text{atan}(\tanh(\text{M6}/\text{b})))$$

$$\text{AHDI} \sim \alpha + \beta(\text{a} * \tan(\tanh(\text{M6}/\text{b})))$$

$$\text{AHDI} \sim \alpha + \beta(\text{a} * \tanh(\text{b} * \text{M6}))$$

$$\text{AHDI} \sim \alpha + \beta(\text{a} * \text{atan}(\text{b} * \text{M6}))$$

$$\text{AHDI} \sim \alpha + \beta(\ (\text{a}/(\text{b} + \exp(\text{-M6}))))$$

$$\text{AHDI} \sim \alpha + \beta((1 + \exp(\text{-M6}))\hat{}(\text{-b})) \qquad (7)$$

$$\text{AHDI} \sim \alpha + \beta(\text{a} * \text{M6}/\text{sqrt}(\text{b} + \text{M6} * \text{M6}))$$

The strongest correlation was achieved by modelling the expression (7), obtaining the final AHDIm expression (8):

$$\text{AHDIm} = \text{-}6.329 + 8.4701(1 + \exp(\text{-M6}))\text{^}(\text{-}0.46443)) \qquad (8)$$

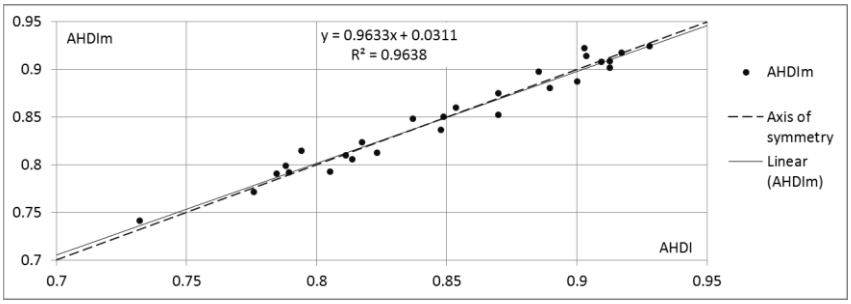

Figure 6. Regularity of the final non-linear model AHDIm vs AHDI. Developed by authors.

One can see that the non-linear model AHDIm (fig. 6) is much more coinciding with the actual AHDI values in comparison with the linear one. All numerical characteristics have been improved (table 2). Residuals of only three countries are over 1.5%. The formula of the linear trendline shows a reduced deviation of the data points from the axis of symmetry. R diagnostic plots (fig 7) show better behavior of the residuals; they are more equally spread around the horizontal zero line (fig. 7a). No data point is even near the *Cook's distance* (fig. 7b).

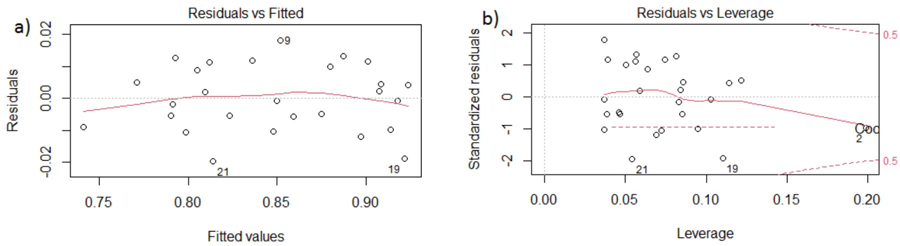

Figure 7. Residual characteristics of the non-linear model. Developed by authors.

4. Results and discussion

The achieved accuracy of the model AHDIm and its coincidence with the real AHDI clearly demonstrates correctness of the trend of the current study and its perspective. The analysis of results (fig. 6) shows that all 27 data points lay in the corridor +/- 2.5%, 24 of them in the corridor +/- 1.5% (see table 2). The quality of the obtained modelling result, which is confirmed by the excellent correlation with AHDI and the statistical significance of all predictors, gives confidence that these six mathematically selected predictors can be defined as SD KPIs.

In order to more accurately reflect the reality, the statistical raw data, which were used as predictors, typically are repeatedly revised and updated by data holders. We checked if and how these stochastic data changes affect the model AHDIm, resp. is the model *stable* against such data variability. Several control models AHDIm5% were computed using the modified data, choosing 5% as the maximum level of the random input data variability; 5% is close to the median change in EU27 data over the previous 3 years. Of course, these changes have a corresponding effect on the modelled assessment of each individual country, however, the shift using models AHDIm and AHDIm5% is negligible; it does not exceed 0.2% across the EU27countries. It means that for practical applications there are not necessary also the annual calibrations of the model AHDIm.

The results obtained could be applied to the cities of the EU27 countries, because: (1) benchmarking modelling was performed in the relatively homogeneous territory of the EU27 countries and cities; (2) all six KPIs are neutral in relation to the scale of the issue covered and (3) KPIs at least in part relate to the competence of the city municipality. It means:

$$\text{AHDIm} > \text{CSDI} \quad (9)$$

When transferring the results at the city level, these KPIs show the priorities for the municipality of the EU city, while the changes of the CSDI show the progress achieved. At the same time it should be strongly mentioned that the values of variables are limited by their indicators in the EU27 countries. This is another reason for the limitation at the EU27 territory.

As a pilot project CSDI values was calculated for several Latvian cities (fig. 8); their population ranges from 20 thousand to 620 thousand. The widespread perception of the highest level of sustainability of the capital Riga among the country's cities is not confirmed. Valmiera (23 thousand inhabitants) and Ventspils (34 thousand inhabitants) have received the highest scores, clearly indicating the sustainability resources in the smaller cities.

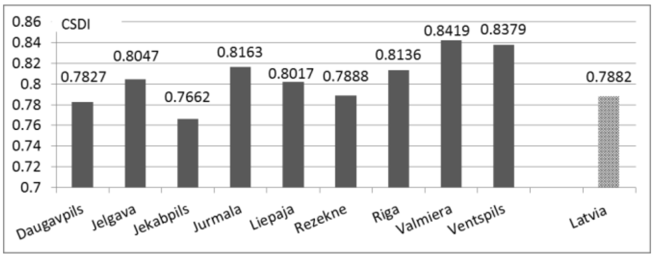

Figure 8. CSDI for cities of Latvia. Developed by authors.

5. Conclusions and recommendations

The concept of the SD defined by the UN in parallel with ongoing urbanization clearly demonstrates the rapidly growing role of cities in global development in the 21st century. This poses huge sustainability challenges and responsibilities for cities municipalities. The efficient implementation of the SD requires regular monitoring / measurement of development progress in the particular city to monitor the achieved level in statics and dynamics, as well as to make strategic decisions for the next period.

The nationally achieved SD level, calculated according to the UN concept, is well characterized by the AHDI – a simple balanced index that is based on result-oriented headline indicators. The AHDI methodology is not directly applicable at the urban level, as it is not possible to determine headline indicators (e.g., Gross National Income) for the individual city.

At the level of EU countries, a relatively large amount of data is available, including sectoral SD indicators, which do not depend on the size of the country, and the progress of which can be more or less influenced and driven by local governments. A working hypothesis was put forward that these indicators could be considered as potential drivers of the SD. Applying adapted data mining algorithms, the most significant SD drivers (KPIs) were selected, their optimal proportions in the complex SD index were mathematically computed and the SD model AHDIm was created. Despite of the usage only six KPIs, the compliance of the model with the actual AHDI can be evaluated as excellent.

As EU countries operate in a harmonious environment (political, regulatory, environmental, etc.) with their cities, the model can be transformed at the urban level, obtaining a simple indicator CSDI of the SD level for any EU city including small and medium-sized ones. The selected methodology allows reflect the results achieved in the implementation of the *Agenda 2030* vision, including taking into account the mutual relationships and inter-impact of the development dimensions. In addition, consistency has been obtained between the SD ratings of the country and its cities.

It has to be strongly mentioned that the KPIs, which are drivers of the model, is a peak of the pyramid. In fact, they reflect the quality of life of the urban community in the broad sense of the term, integrating many aspects.

Mathematical calculation instead of traditional voluntarily preference of some indicators, the reasonable combination of accuracy, stability and simplicity of the model are strong advantages for the practical applications. The performed stability test confirms that annual calibration of the model is not necessary.

Using KPIs values for your city and performing a described computation of the CSDI, the municipality sees an integrated objective assessment of its functioning, while individual KPIs show both success stories and lagging segments that require resource concentration. Comparison with neighbors provides an opportunity to adopt the best practices on the way towards the data-driven SD of the city.

City government can monitor the index and take it into account, making policy decisions and performing programming process of the municipal budget. CSDI can be compared with corresponding index of country and of other cities. Its interpretation is simple: change of any KPI lead to improvement or worsening of sustainability index CSDI.

It should be noted that the municipality can also use the tool to manage the sustainable development of rural areas within the county and/or the whole county or the region. The tool is scale-neutral and applies to any area for which the required KPIs data are available.

The discovered KPIs will help municipalities to find reasonably balanced vectors in the use of its usually limited resources between regulated autonomous and delegated competences. An objective comparative assessment of the proposed development projects (initiatives, actions) in various sectors becomes possible. To promote the city's SD progress, any project should contribute increasing one or more KPIs thus growing the computed SD level by ΔCSDI. The efficiency of the project EF in relation to the urban SD is:

$$EF = \Delta CSDI/Co$$

where: Co – estimated costs of the project implementation.

At the beginning of the municipal budgeting process, typically number of projects is proposed; costs of the total project portfolio usually exceed the available fiscal space for these projects. It can be recommended the selection of the projects to be implemented, using the project efficiency EF as a criterion.

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Appendix. Selected indicators for the modelling of the urban SD.

Designation		*Indicator*	*Source*
SDG3	**G3.1**	**Share of people with good or very good perceived health, % of population aged 16 or over**	Eurostat
	G3.2	Obesity rate by body mass index (BMI), % of population aged 18 or over	Eurostat
	G3.3	Self-reported unmet needs for medical examination and care, % of population aged 16 or over	Eurostat
	G3.4	Number of generalist medical practitioners, per 100 000 population	Eurostat
	G3.5	Neonatal mortality rate, per 1000 live births	WB
	G3.6	Under-five mortality rate, per 1000 live births	WB
	G3.7	Avoidable mortality, per 100 000 population below 75 years	Eurostat
SDG4	G4.1	Early leavers from education and training, % of population aged 18 to 24	Eurostat
	G4.2	Share of top performers in at least one subject (level 5 or 6)	OECD
	G4.3	Share of population with tertiary education (levels 5-8 ISCED 2011) from 15 to 64 years	Eurostat
	G4.4	Adult participation rate in education and training (last 12 months), from 25 to 64 years	Eurostat
	G4.5	Graduates in tertiary education, in science, math., computing, engineering, manufacturing, construction per 1000 of population aged 20-29	Eurostat
SDG5	G5.1	Gender pay gap in unadjusted form, % of average gross hourly earnings of men	Eurostat
	G5.2	Gap in higher educational attainment level, females - males, aged 15-64, percentage points	Eurostat
	G5.3	Gender employment gap, males - females, aged 15-64, percentage points	Eurostat
	G5.4	People gap with good or very good perceived health, males - females, population aged 16 or over, percentage points	Eurostat
	G5.5	Seats held by women in national parliament (%)	Eurostat
	G5.6	Percentage of women among elected local level representatives	CEMR
	G5.7	People gap /males-females/ with at least basic digital skills, 16-74 years old	Eurostat
SDG6	G6.1	Population having neither a bath, nor a shower in their dwelling, % of population	Eurostat
	G6.2	**Population not having indoor flushing toilet for the sole use, % of population**	Eurostat
	G6.3	Population connected to at least secondary wastewater treatment, %	Eurostat
SDG7	G7.1	Share of renewable energy in heating and cooling	Eurostat
	G7.2	**Share of population unable to keep home adequately warm by poverty status, %**	Eurostat
SDG8	G8.1	Employment rate, population from 20 to 64 years	Eurostat
	G8.2	Newly employed (percentage of people in current job for 12 months or less, in total employment), from 15 to 64 years	Eurostat
	G8.3	Long-term unemployment rate, % of active population from 15 to 74 years	Eurostat
SDG9	G9.1	Employment in high- and medium-high technology manufacturing and knowledge-intensive services; % of total employment	Eurostat
	G9.2	**Percentage of enterprises who have ERP software package to share information between different functional areas (10+ employees)**	Eurostat
	G9.3	Percentage of the gross value added by industry (B – E, NACE Rev.2)	Eurostat
	G9.4	Percentage of enterprises (10+ employees) having a fast (> 30 Mbit/sec) fixed broadband connection	DAS
	G9.5	Percentage of enterprises that provided any type of training to develop ICT related skills of their personnel	EIS
	G9.6	Percentage of SMEs who introduced at least one product innovation or process innovation either new to the enterprise or new to their market	EIS

SDG9	G9.7	Percentage of SMEs who introduced at least one new organisational innovation or marketing innovation.	EIS
	G9.8	Percentage of SMEs with in-house innovation activities	EIS
	G9.9	Total innovation expenditure as percentage of total turnover of enterprises, excluding intramural and extramural R&D expenditures	EIS
	G9.10	Percentage of SMEs with innovation co-operation activities	EIS
SDG11	G11.1	Population living in a dwelling with a leaking roof, damp walls, floors or foundation or rot in window frames of floor by poverty status, % of population	Eurostat
	G11.2	Young people neither in employment nor in education and training, from 15 to 24 years, percentage	Eurostat
	G11.3	Average number of rooms per person	Eurostat
SDG12	**G12.1**	**Recycling rate of municipal waste, % of generated**	Eurostat
	G12.2	Recycling rate of e-waste, %	Eurostat
SDG13	G13.1	Greenhouse gas emissions, tonnes per capita	Eurostat
	G13.2	Pollution, grime or other environmental problems, % of households	Eurostat
SDG15	G15.1	Share of forest area, % of total land area	Eurostat
SDG16	G16.1	Individuals interacting online with public authorities in the last 12 months, % of total	DAS
	G16.2	**Perceived independence of the justice system, % of population perceived as very good or fairly good**	Euro-barometer
	G16.3	Crime, violence or vandalism in the area, % of households	Eurostat
SDG17	G17.1	Households with fast (> 30 Mbit/sec) fixed broadband connection, % of total	DAS

Sources: CEMR – The Council of European Municipalities and Regions; DAS – the Digital Agenda Scoreboard; EIS – The European Innovation Scoreboard; OECD – The Organisation for Economic Cooperation and Development, WB – The World Bank.

References

1. Purvis, B., Mao, Y., Robinson, D. Three pillars of sustainability: in search of conceptual origins. *Sustainability Science*, Vol. 14(3), 681-695. doi.org/10.1007/s11625-018-0627-5.
2. World Commission on Environment and Development. Our Common Future. Available on-line: https://sustainabledevelopment.un.org/content/documents/5987our-common-future.pdf (accessed on 15 September 2021).
3. UN General Assembly. Transforming our world: the 2030 Agenda for Sustainable Development. Available on-line: https://www.un.org/ga/search/view_doc.asp?symbol=A/RES/70/1&Lang=E (accessed on 15 September 2021).
4. Meusburger, P. The Spatial Concentration of Knowledge: Some Theoretical Considerations. *Erdkunde,* Vol. 54(4), 352-364. DOI: 10.3112/erdkunde.2000.04.05.
5. Qi, H., Liu, S., Qi, W., Liu, Z. Geographical Concentration of Knowledge- and Technology-Intensive Industries and City Innovation in China. *Sustainability*, Vol. 11(18), 4840. doi.org/10.3390/su11184840.
6. Smart City Hub. Smart cities: facts and figures, drivers and solutions. Available on-line: https://smartcityhub.com/governance-economy/smart-cities-facts-and-figures-drivers-and-solutions (accessed on 15 September 2021).
7. UN Department of Economic and Social Affairs. Sustainable cities and human settlements. Available on-line: https://sustainabledevelopment.un.org/topics/sustainablecities (accessed on 15 September 2021).
8. United Nations Department of Economic and Social Affairs, Division for Sustainable Development. Measuring Progress. In: Prototype Global Sustainable Development Report. Available on-line: http://sustainabledevelopment.un.org/globalsdreport/2014 (accessed on 15 September 2021).
9. Sessa, K. Urban sustainability indicator systems in the EU. In: Bina, O. et al (eds.). *State of the Art Report: Sustainability dimension of urban infrastructure and services.* URBACHINA Project Report no. D4.1, Instituto de Ciencias Sociais (ICS), Institute of Studies for the Integration of Systems (ISIS), and People's University (RENDA), Lisbon. 99-107. Available on-line: https://www.researchgate.net/publication/257125530_State_of_the_Art_Report_Sustainability_dimension_of_urban_infrastructure_and_services (accessed on 15 September 2021).
10. Sharifi, A. A Typology of Smart City Assessment Tools and Indicator Sets. *Sustainable Cities and Society*, Vol. 53, 101936. doi.org/10.1016/j.scs.2019.101936.

11. Sustainable Development Solutions Network. European Cities SDG Index; prototype version. Available on-line: https://euro-cities.sdgindex.org/#/ (accessed on 15 September 2021).
12. Yakoub, W. A., et. al. Urbanism as it should be: an overview of Environmental Urban Movements and Initiatives towards Sustainability. Second International Conference (Tenth Conference of Sustainable Environmental Development) Available on-line: https://www.researchgate.net/publication/332159023_Urbanism_as_it_should_be_An_overview_of_Environmental_Urban_Movements_and_Initiatives_towards_Sustainability (assessed on 15 September 2021).
13. Simon D. Sustainable cities in sustainable societies. In: Simon, D. (ed.). *Rethinking Sustainable Cities; Accessible, green and fair*. Policy Press, University of Bristol, Bristol. 1-10. Available on-line: https://library.oapen.org/handle/20.500.12657/32166 (assessed on 15 September 2021).
14. Arcadis. Citizen Centric Cities; the Sustainable Cities Index. Available on-line: https://www.arcadis.com/media/1/D/5/%7B1D5AE7E2-A348-4B6E-B1D7-6D94FA7D7567%7DSustainable_Cities_Index_2018_Arcadis.pdf (assessed on 15 September 2021).
15. Bee smart city. Building the Smart City: Advancing in Six Fields of Action. Available on-line: https://hub.beesmart.city/en/smart-city-indicators (assessed on 15 September 2021).
16. CITYkeys. CITYkeys indicators for smart city projects and smart cities. Available on-line: https://nws.eurocities.eu/MediaShell/media/CITYkeysD14Indicatorsforsmartcityprojectsandsmartcities.pdf (assessed on 15 September 2021).
17. IMD. IMD Smart City Index. Available on-line: https://www.imd.org/research-knowledge/reports/imd-smart-city-index-2019/ (assessed on 15 September 2021).
18. Sasanpour, F. Liveable City; One Step towards Sustainable Development. *Journal of Contemporary Urban Affairs*, Vol. 1(3), 13-17. doi.org/10.25034/ijcua.2018.3673.
19. The Economist Intelligence Unit. The Global Liveability Index 2021. Available on-line: https://pages.eiu.com/rs/753-RIQ-438/images/global-liveability-index-2021-free-report.pdf?mkt_tok=NzUzLVJJUS00MzgAAAF9j7zv1yd6L0CbgW-lymzHYZWBK0zM4dUpS7T5wmBllfMLvM9-iEEV1aEhHr6WWX1c5H_3uIjZ2JJFoUTIrR0vIh8UJxcLAQBppVaItXdYRQM (assessed on 15 September 2021).
20. Bulkeley, H., Marvin, S. Urban Governance and Eco-cities: Dynamics, Drivers and Emerging Lessons. In: Hofmeister, W., Rueppel, P., Fook, L. L. (eds.). Eco-Cities: Sharing European and Asian Best Practices and Experiences. Singapore, European Union Centre; 19-34.
21. Karnitis, E., Bicevskis, J., Karnitis, G. Measuring the Implementation of Agenda 2030 Vision in its Comprehensive Sense: Methodology and Tool. Energies 2021, Vol. 14(4), 856. doi.org/10.3390/en14040856.
22. Karnitis, E., et al. Action Lines of Municipalities to Improve the Sustainable Development of Small and Medium-sized Cities in the EU. J. of Int. Sci. Publications; Economy & Business, vol. 15. In publishing.
23. Moriarty, J. P. A Theory of Benchmarking, Benchmarking: An International Journal, Vol. 18(4): 588-611. doi.org/10.1108/14635771111147650.
24. Karnitis, G., Karnitis, E.: Sustainable Growth of EU Economies and Baltic Context: Characteristics and Modelling. J. Int. Studies, Vol. 10(1), 209-224. DOI: 10.14254/2071-8330.2017/10-1/15.
25. Karnitis, G., Virtmanis, A., Karnitis, E. Key Drivers of Digitalization; EU Context and Baltic Case. Baltic J. Modern Computing, Vol. 7(1), 70-85. DOI: 10.22364/bjmc.2018.7.1.06.